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To cite this article: Yifan Zhang, Wenbo Zhang, Ziyi Zeng, Keying Jiang, Jingxuan Li, Wen Min, Wei Luo, Qingfeng Guan, Jianfeng Lin & Wenhao Yu (2025) MapReader: a framework for learning a visual language model for map analysis, International Journal of Geographical Information Science, 39:9, 1973-2008, DOI: [10.1080/13658816.2025.2455112](https://doi.org/10.1080/13658816.2025.2455112)

To link to this article: <https://doi.org/10.1080/13658816.2025.2455112>



Published online: 28 Jan 2025.



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RESEARCH ARTICLE



MapReader: a framework for learning a visual language model for map analysis

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ABSTRACT

Intelligent map analysis is an important yet challenging topic. Recently, the development of large models, especially Visual Language Models (VLMs), has shown potential for intelligent image analysis. However, these models are primarily trained on natural images, which have intrinsic differences from maps. Consequently, there remains a gap in applying existing general-domain VLMs to map analysis. To address this issue, we propose a framework for developing a specialized VLM, called MapReader. To achieve this goal, a comprehensive data resource is collected using a strategy that combines self-instruct with expert refinement, including training data (MapTrain: 2,000 pairs of maps and descriptions) and evaluation data (MapEval: 250 maps and 500 map-related questions). Based on the training data, MapReader is fine-tuned on top of a general-domain VLM to learn to understand and describe map contents. The evaluation results on MapEval suggest that: (1) MapReader can accept map inputs and generate detailed descriptions of core geographic information, and it also possesses visual question-answering capabilities, showing potential for application in various map analysis scenarios, such as accessible map reading and robotic map usage; (2) The proposed data collection strategy is effective, and the collected dataset can serve as a benchmark to promote further map analysis research.

ARTICLE HISTORY

Received 25 August 2024

Accepted 13 January 2025

KEYWORDS

Map analysis; map understanding; cartography; visual language model

1. Introduction

With advancements in surveying and communication technologies, the variety and styles of maps have expanded significantly, making them more widely used and enhancing their importance across various fields (Lee 1995, Peterson 1997, Veenendaal *et al.* 2017, Savino *et al.* 2021). However, despite the widespread availability of maps, readers without formal map-reading skills may struggle with map

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analysis and usage (Davis 1973, Ooms *et al.* 2016, Carbonell Carrera *et al.* 2017, Zhang *et al.* 2024b). These struggles can result in navigation errors and reduced efficiency in both personal and professional contexts. For instance, in emergency response scenarios (e.g. disaster response), misinterpretation of map information by untrained personnel can lead to delayed response times and potentially compromised public safety outcomes. Researchers have been attempting to solve this problem from different perspectives.

On the one hand, from the cognitive perspective, some researchers investigated the factors that influence the map analysis process of humans (Gerber 1984, Lobben 2004, Ishikawa 2021). Additionally, some researchers focused on standardized symbols and simplified map formats to help non-experts interpret spatial data accurately (MacEachren 2004, Krygier and Wood 2016). Moreover, many institutions also emphasized accessible map design to improve comprehension (e.g. FGDC Digital Cartographic Standard for Geologic Map Symbolization (U.S. Geological Survey 2006)). Such regulations underscore the social responsibility in map-making, aiming to bridge the gap in spatial literacy and mitigate potential societal impacts.

On the other hand, some researchers attempted to explore technologies and methods that can assist users in the map analysis process, enabling them to easily gain geographic information from maps (Uhl and Duan 2020, Kim *et al.* 2023). For instance, some rule-based methods have been designed to identify specific objects from maps (e.g. textual annotations (Pezeshk and Tutwiler 2011) and contour lines (Khotanzad and Zink 2003)). With the advent of deep learning, some researchers have explored using deep learning models to aid map analysis (Kang *et al.* 2024), including the extraction of different types of geographic objects (Hussain and Shan 2016, Maxwell *et al.* 2020, Jiao *et al.* 2021, Vynikal *et al.* 2024). However, these methods are typically designed or trained for specific types of geographic objects, and often lack the capability to identify other objects. More importantly, the identified geographical objects by these methods are still presented in visual form, which restricts their functions to assist users in obtaining geographical information.

To facilitate easier comprehension of map content, some researchers have explored more direct map analysis ways by outputting map contents in natural language form. For example, Li *et al.* (2018) proposed a three-tiered framework that first extracts textual content from maps, then recognizes the text, and finally constructs a dictionary for matching the extracted text to output map content. However, this method primarily supports the extraction of textual annotations and relies heavily on the construction of dictionaries, limiting its vocabulary range of output. Therefore, exploring a feasible method or model that can analyze maps effectively and output focused geographic information in natural language is necessary.

To solve this problem, this paper proposes a framework for intelligent map analysis, which can generate natural language descriptions of map content. Our approach is inspired by existing Visual Language Models (VLMs). In computer vision, analyzing image content has been a longstanding research focus, evolving from image classification (Figure 1(a)) (Rawat and Wang 2017) and semantic segmentation (Figure 1(b))

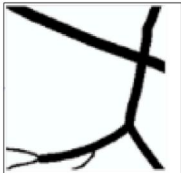
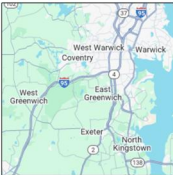
Natural image		Map	
	⇒ Car ✓ Bus Train Plane Ship		⇒ Urban theme ✓ Traffic theme Statistical theme Indoor theme ...
(a) Image classification (Rawat and Wang, 2017)		(d) Map-type classification (Wen et al. 2024)	
			
(b) Image semantic segmentation (Guo et al. 2018)		(e) Map object extraction (Jiao et al. 2022)	
	⇒ The image showcases a railway platform labeled 'Platform 2'. A green train is approaching, with steam emanating from its engine. On the platform, there are ...		⇒ The map depicts a specific region, highlighting several towns and major roads. The towns of West Warwick ... are labeled ... Water bodies are present in the eastern part of the map, near Warwick ...
(c) Image analysis (Wang et al. 2023)		(f) Map analysis (Our focus)	

Figure 1. With the development of computer vision, such as (a) image classification, (b) semantic segmentation, and (c) visual question answering, many researchers have also designed relevant models for assisting users in extracting geographic information from maps, including (d) map-type classification and (e) map object extraction. In this paper, our focus is on map analysis (f), which aims to enable a model to understand maps and accurately describe map content, including geographic objects and their corresponding spatial relationships.

(Guo et al. 2018) to visual question answering (Figure 1(c)) (Wang et al. 2023). These models can describe the contents of an image or answer user queries based on the image content. Motivated by these developments in computer vision, map researchers have begun applying visual models to map-related tasks, such as map-type classification (Figure 1(d)) (Wen et al. 2024) and map object extraction (Figure 1(e)) (Jiao et al. 2022). Recently, influenced by advancements in Large Language Models (LLMs) and multimodal models, VLMs have gained significant attention (Zhang et al. 2024a). However, there has been no significant effort to introduce VLMs into cartography. The main challenge lies in the differences between maps and natural images.

Specifically, maps and natural images are distinct data representations. Maps abstractly represent the real world, contain high-dimensional geographic information, and it require specialized geographic knowledge for map analysis (Vasiliev et al. 1990). Generally, there are a set of professional paradigms designed to convey geographic information within maps, such as map styles, layout, legend, scale, and map symbols (MacEachren 2004). In contrast, natural images are relatively realistic representations of

the real world, which capture the appearance of objects or scenes as they appear to the eye. Therefore, applying general-domain VLMs trained mostly on natural images to map analysis tasks may not obtain satisfactory results.

1.1. Research purpose

Therefore, in this paper, we propose a framework for learning a professional VLM, called MapReader, for intelligent map analysis (Figure 1(f)). To achieve this goal, a comprehensive data resource is first collected, including training data (MapTrain) and evaluation data (MapEval). Labeling map content in text requires specialized knowledge, and relying on manual annotation of map features demands significant human resources. Therefore, this paper employs a strategy that combines self-instruct with expert refinement. Specifically, initial map descriptions are annotated by leveraging one of the most powerful VLMs, GPT-4o; the results are then reviewed and optimized by cartographic experts. In this way, a total of 2,000 pairs of maps from various geographic contexts and their corresponding descriptions are obtained to construct the training data MapTrain. Then, the evaluation data MapEval is constructed from two perspectives to assess the model's performance: the accuracy in describing geographic objects and their spatial relationships within maps (MapEval-DB, 250 map cases), and the effectiveness in answering given questions based on specific maps (MapEval-QA, 500 map related questions). Specifically, MapEval-DB (describe) requires MapReader to describe the core content in given maps with different scales and various contexts, and MapEval-QA (question answering) is used to test whether MapReader can answer user questions considering specific map content.

1.2. Research contribution

Since it requires high computing resources to train a VLM from scratch, MapReader is fine-tuned on top of a general-domain visual language foundation model, CogVLM, to learn to understand and describe map contents. During the training process, MapReader accepts map inputs and is required to output detailed descriptions of the core geographic information within them. Through extensive experiments on the evaluation data, MapReader demonstrates significant improvements in identifying geographic objects and corresponding spatial relationships and providing more detailed map content descriptions than general VLMs. Moreover, MapReader has visual question-answering capabilities, allowing it to answer users' focused questions based on the input maps. The experimental results validate the effectiveness of this framework, and the proposed MapReader has the potential to be applied to various scenarios, such as accessible map reading and robotic map usage (Blades *et al.* 1999, Perkins and Gardiner 2003).

The contribution of this paper can be summarized as follows:

- First, we propose a framework for automatic map analysis, which can be used to assist users in easily understanding maps and obtaining geographical information within them.

- Second, unlike traditional map analysis methods that are designed for single tasks, the proposed MapReader supports multiple tasks, such as map analysis and visual semantic question answering.
- Finally, given that most outstanding VLMs are closed-sourced (e.g. GPT-4o), we collect a professional dataset MapTrain that can be used to enhance the map analysis ability of open-sourced general VLMs. Moreover, our collected evaluation data MapEval can also serve as a benchmark to promote the development of professional VLMs in the cartography community.

2. Related work

How to effectively obtain geographic information from maps is a critical problem (Phillips 1979, McKenzie and Klippel 2016). Generally, relevant strategies or methods are designed from two perspectives to solve this problem. The first aspect focuses on intuitive map design to make it easier for users to understand and retrieve geographic information from maps (Streeter *et al.* 1985). This includes designing appropriate map symbols and legends for clear information presentation (Robinson *et al.* 2012, 2011), optimizing label placement using map label placement algorithms (Zoraster 1986, Barrault 2001, Lei *et al.* 2021), and employing map generalization techniques to ensure clear and legible map content (Brassel and Weibel 1988, Mackaness and Beard 1993, Touya *et al.* 2019). Then, from another aspect, developing models and methods that assist in map comprehension by extracting geographic information from maps is also focused, which can reduce the difficulty of map reading, and aid users in understanding geographic information (Tanaka and Ichikawa 1988, Li and Qi 2003, Hiramoto and Sumiya 2006). Our focus is on the second perspective.

As the need for intelligent map analysis has grown significantly, Researchers have proposed various methods to assist users in map understanding. Traditional approaches include rule-based algorithms, such as template matching and edge detection, for classifying geographic features in map data that exhibit specific characteristics (Yu *et al.* 2017, Chen *et al.* 2021). With the advancement of deep learning, researchers have also explored its application in map analysis, such as designing map classification algorithms to automatically categorize maps into different types (e.g. urban maps, rural maps, transportation maps, and topographic maps) (Wen *et al.* 2024). Additionally, object recognition and extraction algorithms have also been developed to extract specific geographic features of interest from maps (Li *et al.* 2000, Biswas and Das 2012, Feng *et al.* 2020, Xu *et al.* 2023). Kang *et al.* (2024) summarized literature on GeoAI-based map analysis.

However, given a map, users may want to obtain more geographical information than only identified geographic objects. Specifically, it is also important to understand the spatial relationships between them (Chen *et al.* 2007, Steiniger and Weibel 2007). However, compared to object recognition on maps, this is a finer-scale and more challenging task to convey information about both the geographic objects and corresponding spatial relationships on maps to users. Therefore, designing methods or models to provide fine-scale map content information for

users is a focus in the cartography domain (Tlauka *et al.* 2005, Levit and Roy 2007, Meneghetti and Pazzaglia 2021).

In recent years, with the development of large models, particularly the commercial application of LLMs like ChatGPT, researchers have begun to explore whether large models can be used to solve specialized tasks (Mai *et al.* 2024, Wei *et al.* 2024, Zhang *et al.* 2024e). For example, in computer vision tasks, previous deep learning models focused on extracting increasingly finer details from images, evolving from image classification (at the image level) to semantic segmentation (at the pixel level) (Wiley and Lucas 2018). Recently proposed VLMs further extend the scope of extracted image information (Du *et al.* 2022). Specifically, given an image, VLMs can describe the detailed information in text format. In other words, with a robust VLM, users can obtain almost all the relevant image information without actually viewing the image.

Inspired by these studies, we explore the potential of using VLMs to assist in map analysis. However, current general-domain VLMs are primarily trained on extensive natural image data, resulting in a lack of specialized understanding of maps. Therefore, our focus is on training a professional VLM specifically for map analysis. Training a VLM in specialized domains presents two major challenges: the availability of computing resources and the corresponding training data (Dou *et al.* 2022, Chen *et al.* 2023). Due to the large parameter size of VLMs, training them demands significant computing resources, such as GPU memory. Additionally, acquiring appropriate training data for specialized domains is crucial. To address the first challenge, researchers have proposed lightweight fine-tuning strategies, such as LoRA (Hu *et al.* 2021) and adapters (Yang *et al.* 2024). These strategies have been successfully applied in adapting large pre-trained models to specialized domains, yielding satisfactory results (Duan *et al.* 2024, Zhang *et al.* 2024c). However, the second challenge remains—the difficulty in creating map-description pair data has constrained further research in this area. Unlike natural image descriptions, map analysis—encompassing geographic object identification and spatial relationship analysis—requires expertise, which limits the ability to collect large-scale data. To overcome this issue, our study proposes a strategy that combines self-instruct with expert input to collect map-description pair data for fine-tuning VLMs. The corresponding data collection and training methods will be detailed in Sections 3 and 4.

3. MapReader data collection and processing

In this paper, we collect a comprehensive data resource, including training data (MapTrain, Section 3.1) and evaluation data (MapEval, Section 3.2). Specifically, the training data, MapTrain, is collected to fine-tune a general-domain VLM, CogVLM Wang *et al.* (2023), to adapt it to map data. Subsequently, the evaluation data, MapEval, is constructed to assess the fine-tuned model from two perspectives: its ability to accurately describe the content of given maps and its capability to answer visual questions based on specific maps. The data collection process involves data downloading, filtering, and labeling. The framework used to collect MapTrain and MapEval is presented in Figure 2, with details elaborated in this section.

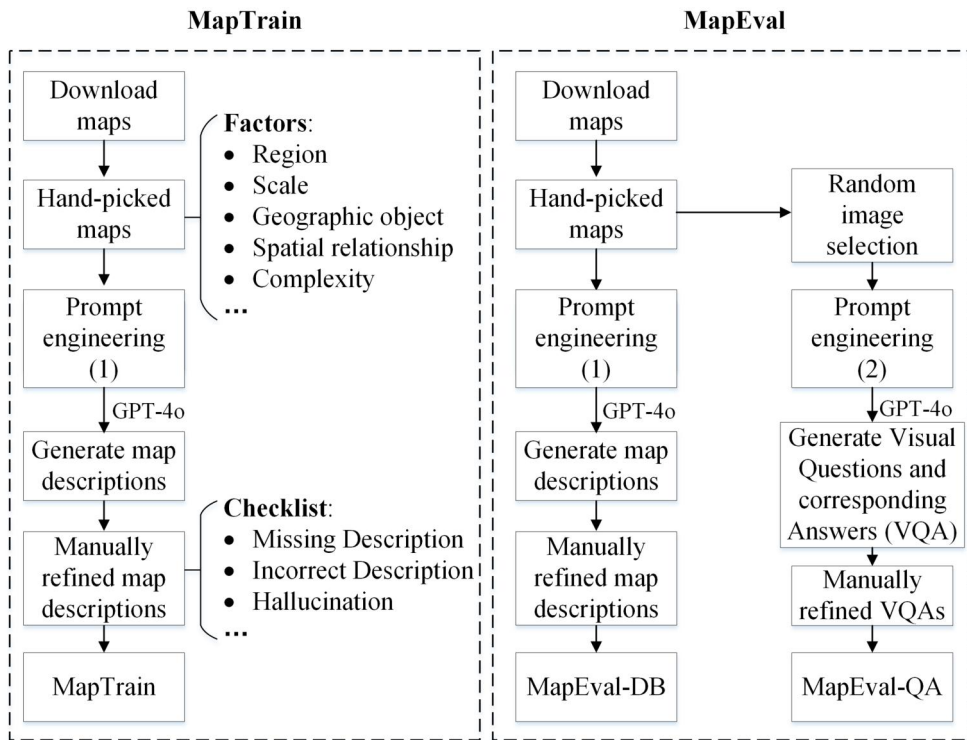


Figure 2. The framework to construction training data MapTrain and evaluation data MapEval.

3.1. Training dataset: MapTrain

From Figure 2, two major processes are involved in collecting MapTrain. The first is Map collection (Section 3.1.1), which includes the steps of Downloading maps and Hand-picking maps; the second is Map description generation (Section 3.1.2), which includes the steps of Prompt engineering, Generating map descriptions, and Manually refining map descriptions. To fine-tune a VLM for a specific domain, the quality of training data is important, primarily from two perspectives: the quality of visual images (V-perspective) and the quality of textual descriptions (L-perspective). In this section, the process of collecting MapTrain is elaborated, explaining how we consider both perspectives.

3.1.1. Map collection

From the V-perspective, multiple factors are considered to collect a high-quality set of maps. First, the richness of geographic contexts within maps is crucial, as it affects the ability of models trained on this dataset to generalize across various geographic scenarios. Therefore, maps in MapTrain are downloaded from different regions within the New York and Providence areas in the United States, which ensures the inclusion of maps with diverse geographic features and contexts, such as road networks, buildings, lakes, and railways, to name a few. Additionally, since map scale is also one of the key elements, maps with different scales are obtained. In this paper, the maps are downloaded from Google Maps, with zoom levels

ranging from 10 to 17, corresponding to map scales from 1:800,000 to 1:2,300. Some data examples are presented in [Appendix A](#).

With these downloaded maps, factors relevant to detailed map contents are also considered. First, geographic objects are generally presented as different map symbols on maps. In this way, to enable a VLM to effectively recognize different geographic objects, various geographic objects (i.e. map symbols) should exist in the collected training data. Therefore, five cartography experts are required to manually filter to keep maps based on the following criteria: (1) An effective map sample should contain abundant geographic information, such as textural annotations or geographic objects (e.g. buildings, road networks, and POIs); (2) Meanwhile, maps without geographic objects or with blurry or significantly distorted contents will be removed. In this way, the kept maps contain abundant geographic objects, and corresponding spatial relationships will also be presented in them. Finally, a total of 2,000 maps with high quality are obtained in the MapTrain.

3.1.2. Map description generation

From the L-perspective, generating accurate and complete descriptions of map content is important. Given a specific map, the map description should contain all the textural annotations and geographic objects presented on it. Moreover, the descriptions should also capture key spatial relationships among these geographic objects. However, with these criteria, it is labor-consuming to manually label all the maps, since this process requires expert cartographic knowledge and involves multiple steps. Therefore, in this paper, we adopt a strategy that combines self-instruct with expert refinement to generate effective map descriptions.

The self-instruct strategy was first proposed to overcome the constraint that the corpus used to train LLMs heavily relies on manually labeled data (Wang *et al.* 2022). Specifically, given the robust semantic understanding and text generation capabilities of LLMs, this strategy uses a robust LLM (such as GPT-3) to generate questions on diverse topics and also generate corresponding answers. In this way, these questions and answers can be organized as instruction-following data, which can be used to fine-tune other LLMs. Experimental results show that the generated training data can match the effectiveness of manually labeled data, and this strategy has been validated in various fields, such as law (Cui *et al.* 2023), medicine (Thirunavukarasu *et al.* 2023), and GIS (Zhang *et al.* 2024d).

Inspired by this idea, we extend the self-instruct strategy to collect map descriptions. Specifically, a robust commercial VLM, GPT-4o, is used to generate descriptions for maps. GPT-4o is a recently proposed VLM, which presents a robust capability to understand and describe images. Since GPT-4o is trained on a large amount of data, we tested it and found it also capable of understanding map content. To generate effective map descriptions, we designed the following prompt after multiple tests: ‘You are an expert in cartography. Please describe the given map in detail, including detailed information about geographical entities and spatial relationships. Do not make up or fabricate any details’.

With this prompt, GPT-4o is used to generate an initial description for each map, and this process greatly reduces the time required for manual map content

description labeling while providing relatively high-quality initial descriptions. However, since GPT-4o is not a professional VLM designed for cartography, there are still some unsatisfactory map descriptions, such as missing descriptions, incorrect descriptions, and even hallucinations. Therefore, the initial map descriptions are also manually rigorously refined to ensure their professionalism and high quality. The refinement process mainly includes the following steps:

- **Language refinement:** The generated map descriptions are refined to ensure that the language is more concise, accurate, and professional. This improves the readability and efficiency of information delivery. For example, we simplified complex sentence structures into easily understandable short sentences and avoided redundant words.
- **Content refinement:** After language refinement, the descriptions are carefully checked and verified to ensure that the information in the captions is completely consistent with the actual content of the maps. This step is crucial to avoid any potential misinformation. Moreover, missing descriptions of key map elements are also added manually. Each generated map description has been proofread to ensure its accuracy, consistency, and comprehensiveness.
- **Semantic optimization:** Finally, the semantics of the map descriptions are optimized to enhance their clarity and logical coherence, making them easier to understand. For example, the descriptions of geographic names and landmarks are checked to be accurate and use standardized terms.

In the end, we collected 2000 high-quality map-description pairs. An illustrative example of the entire map description generation process is presented in [Figure 3](#). First, for a specific map ([Figure 3\(a\)](#)), the robust VLM GPT-4o generates an initial description with the self-instruct strategy ([Figure 3\(b\)](#)). However, there are often unexpected (i.e. inaccurate) or missing details in the initial description. To address this, an expert refinement process (e.g. adding missing descriptions or revising inaccuracies in [Figure 3\(b\)](#)) is applied to produce the final description ([Figure 3\(c\)](#)).

Therefore, the final description provides rich and specific information about the geographic scene, including geographic objects and their corresponding spatial relationships on the map. These descriptions can provide valuable references for users to interact with the map. With these map-description pairs, the MapTrain can be constructed. This can be used to train a professional VLM that accepts map input and outputs corresponding geographic information, similar to what is presented in [Figure 3](#).

3.1.3. Statistics of MapTrain

In this section, the statistics of MapTrain are presented to provide more information about it. First, the distribution of map scales is shown in [Figure 4](#), which presents that MapTrain contains maps with various scales. Generally, the smaller the map scale, the wider the area described by the map. In any given area, there are fewer maps with smaller scales compared to those with larger scales. Therefore, in MapTrain, maps with larger scales (e.g. zoom levels 15 and 16) are more common than maps with smaller

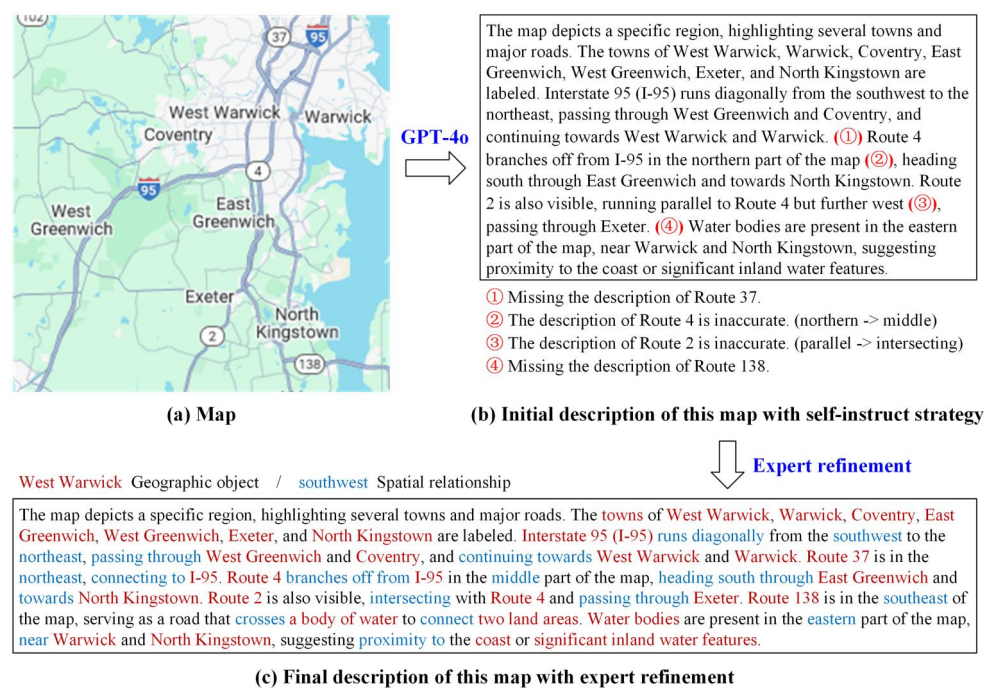


Figure 3. An illustration of the map description generation process. First, given a specific map (a), an initial description is generated using the self-instruct strategy with a robust Visual Language Model, GPT-4o (b). Subsequently, this description is further refined through an expert refinement process to produce the final version (c).

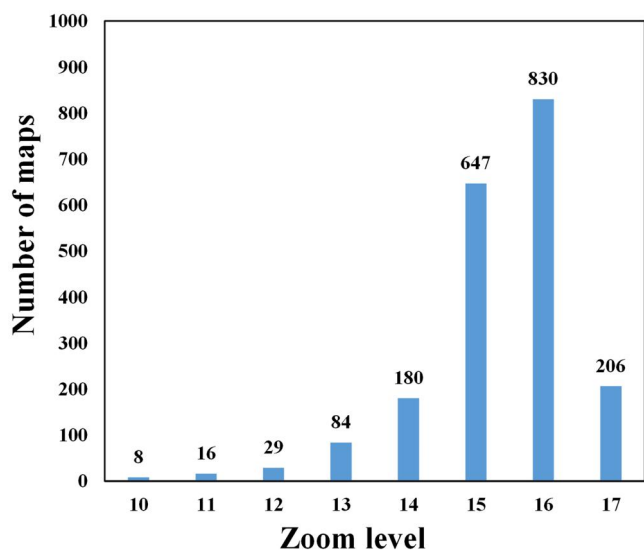


Figure 4. The distribution of map scales in MapTrain, with zoom levels ranging from 10 to 17, corresponds to map scales from 1:800,000 to 1:2,300.

scales (e.g. zoom levels 10, 11, and 12). However, when the map scale is too large, there will be less geographic content because the area covered is too limited, resulting in less rich map descriptions. Thus, to collect a high-quality map-description pair

dataset, there are also fewer maps with zoom level 17 compared to those with zoom levels 15 and 16.

Next, the statistics of map descriptions are presented. The distribution of geographic object-related words and spatial relationship-related words (illustrated in Figure 3) in the description of each map are shown in Figure 5(a,b). According to the statistics in Figure 5, the generated map descriptions vary in complexity, with different numbers of geographic objects and spatial relationships. This variation allows the VLM trained on MapTrain to handle maps with varying levels of complexity.

Finally, to illustrate the geographic contexts in the maps of MapTrain, Figure 6 presents the frequency distribution of geographic object-related and spatial

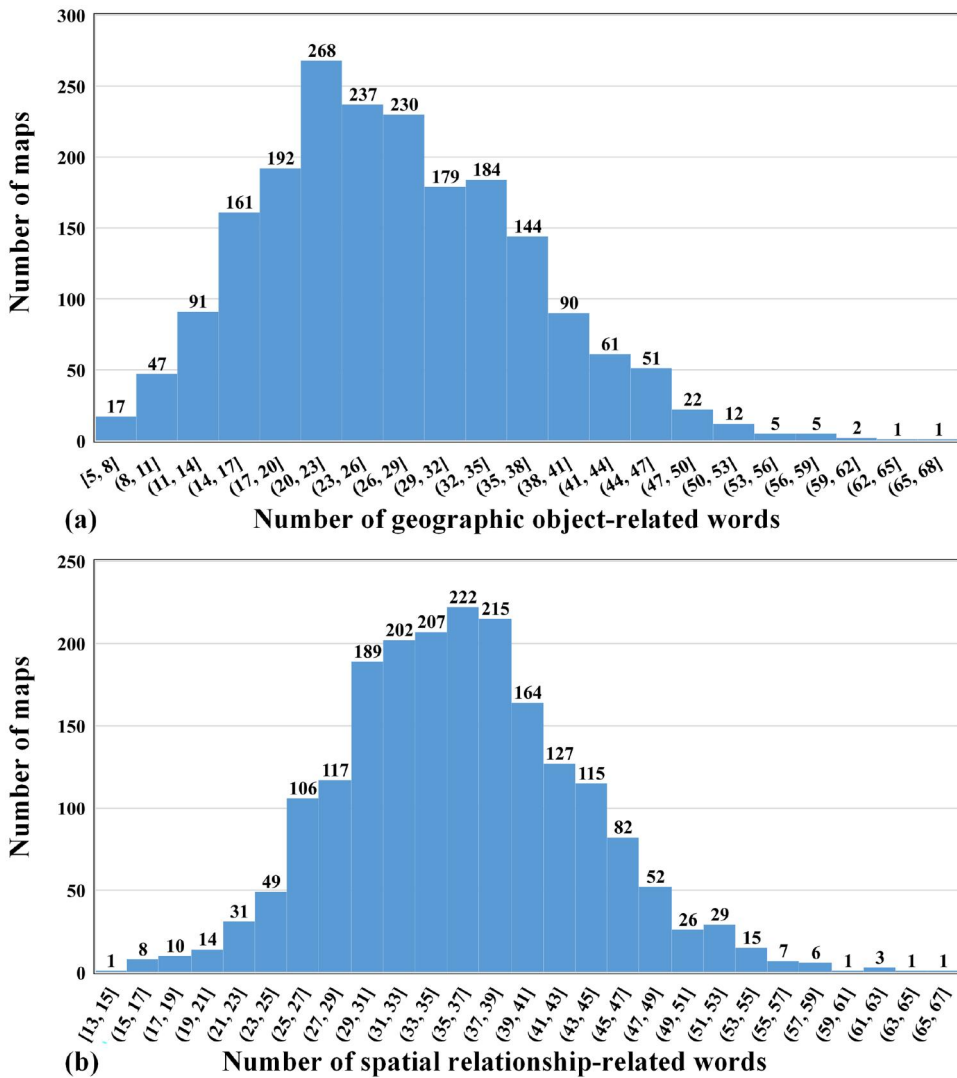


Figure 5. The quantity distributions of (a) geographic object-related words and (b) spatial relationship-related words in map descriptions of MapTrain, with x-axis intervals representing word count ranges (e.g. [5–8] and [13–15] indicate the number of words in that range).

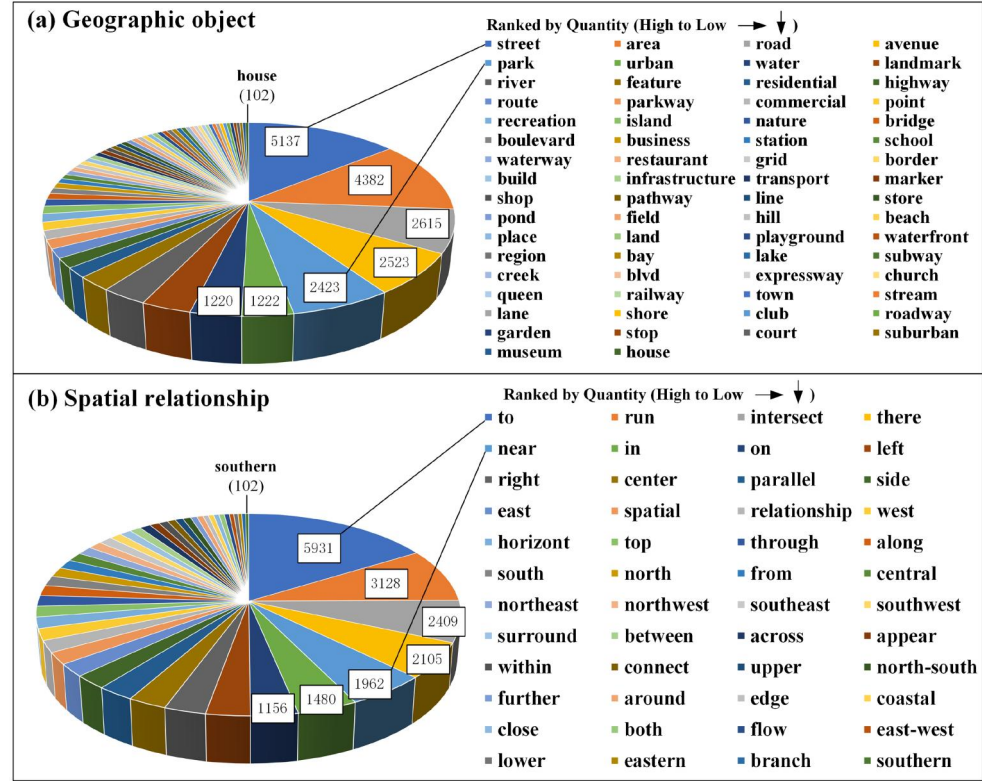


Figure 6. Frequency distribution of (a) geographic object-related and (b) spatial relationship-related words in map descriptions of MapTrain (words with frequency greater than 100).

relationship-related words in the map descriptions of MapTrain. From Figure 6, it can be observed that there are various geographic contexts in MapTrain, which provide abundant geographic information to the trained VLM, enhancing its ability to handle different geographic contexts.

3.2. Evaluation dataset: MapEval

In this paper, we also collect a benchmark to evaluate the performance of VLMs in understanding map content from two perspectives, including the ability to accurately describe the content of given maps (Section 3.2.1) and its capability to answer visual questions based on specific maps (Section 3.2.2).

3.2.1. Describing map: MapEval-DB

To evaluate the performance of VLMs in describing map content, MapEval-DB is collected, which contains maps and their corresponding detailed map descriptions. Similar to the process of collecting MapTrain, MapEval-DB is also collected following the strategy presented in Figure 2. However, to test the ability of the trained VLM on maps across various geographic contexts, maps in MapEval-DB are downloaded from different areas, including urban areas within New York and Orlando (200 maps), as

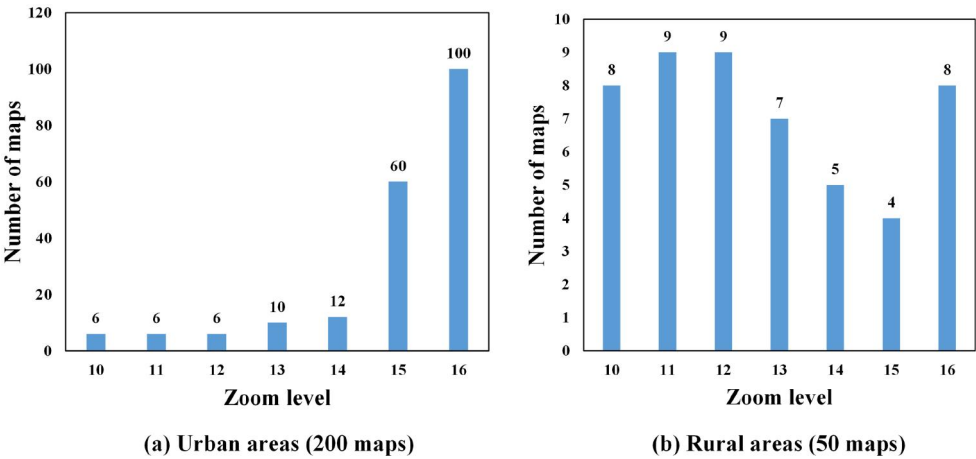


Figure 7. The distribution of map scales in (a) urban areas and (b) rural areas in MapEval-DB, with zoom levels ranging from 10 to 16, corresponds to map scales from 1:800,000 to 1:12,000.

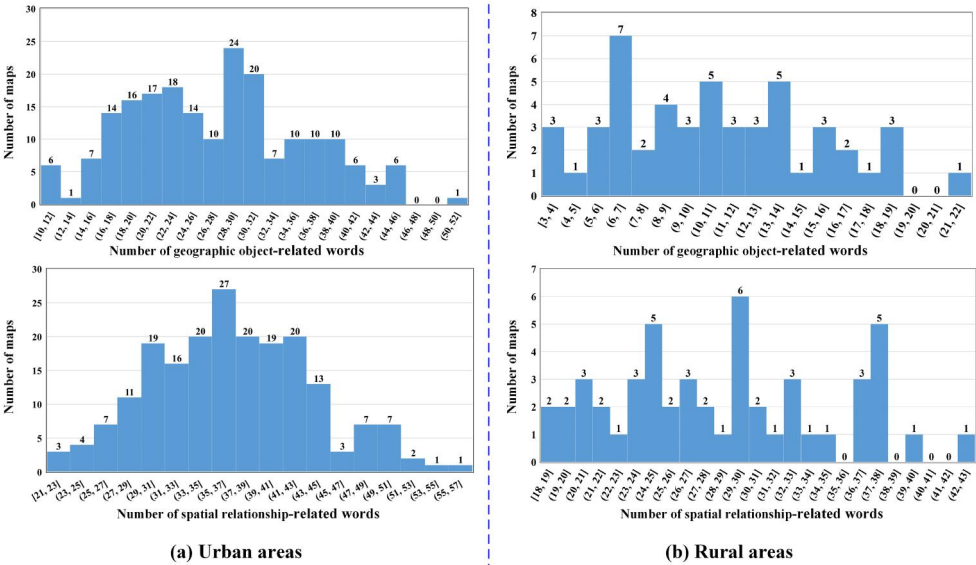
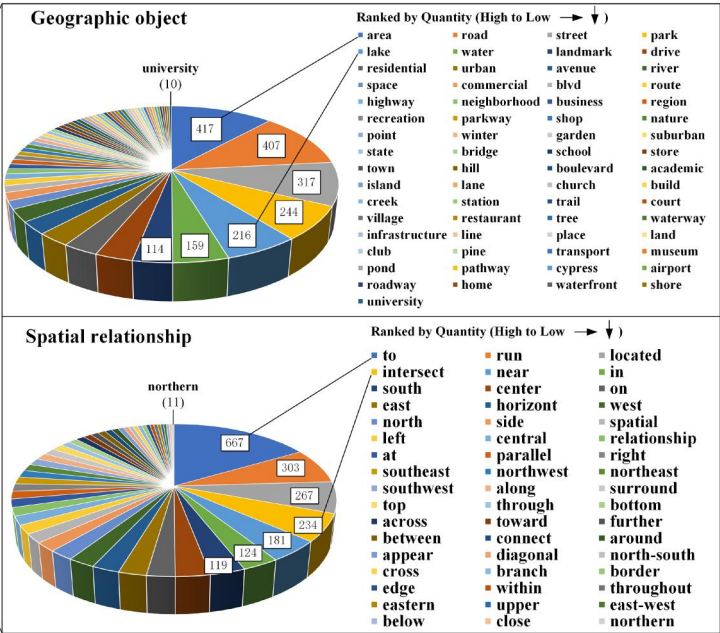


Figure 8. The quantity distributions of geographic object-related words and spatial relationship-related words contained in map descriptions for (a) urban areas and (b) rural areas of MapEval-DB, with x-axis intervals representing word count ranges (e.g. [10–12] and [3–4] indicate the number of words in that range).

well as rural areas from different countries (50 maps). In this way, a total of 250 maps are obtained, and the distribution of map scales is shown in Figure 7. Next, the corresponding map descriptions are also initially generated based on a robust commercial VLM, GPT-4o. After that, each map description will be reviewed and optimized by two experts to ensure its completeness and correctness. After the collection, relevant statistics of MapEval-DB are presented in Figures 8 and 9. From these data distributions, distinct characteristics emerge between the map descriptions for urban and rural

(a) Urban areas



(b) Rural areas

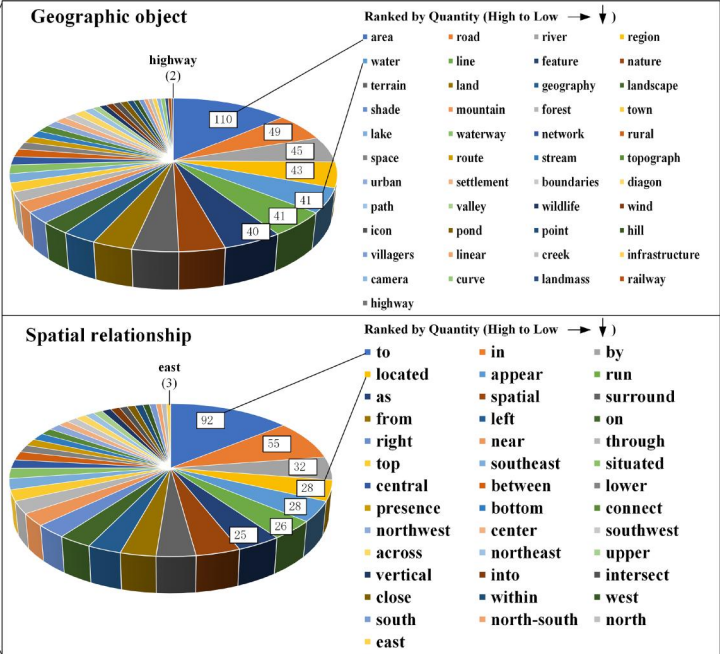


Figure 9. Frequency distribution of geographic object-related and spatial relationship-related words in map descriptions for (a) urban areas and (b) rural areas of MapEval-DB.

areas. Specifically, word quantity is generally higher in urban areas than in rural areas, likely due to the greater number of geographic objects and corresponding spatial relationships present in urban environments. Additionally, the frequently used words differ between the two contexts (e.g. ‘residential’ in urban areas and ‘terrain’ in rural areas).

In summary, MapEval-DB covers maps with various geographic contexts and complexity, making it an effective benchmark that can be used to evaluate VLMs in map analysis.

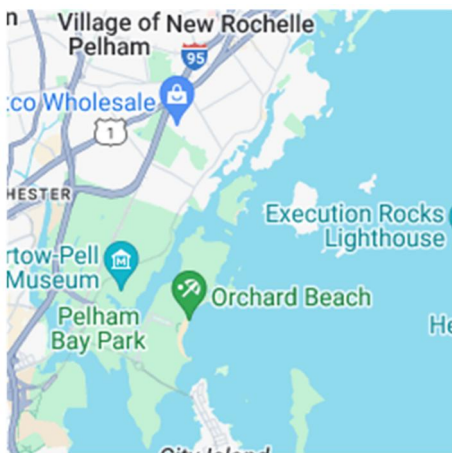
3.2.2. Visual question answering on map: MapEval-QA

For map users, it is common that they may want to ask questions about the geographic information they focused on in a map rather than knowing all the map contents. In this way, a visual question answering benchmark on maps is also collected to evaluate whether the trained VLM has this ability. Specifically, 70 map samples (50 from urban areas and 20 from rural areas) representing different geographic contexts are selected from the MapEval-DB to serve as the base maps. Next, to obtain map-related question-answer pairs, a carefully crafted prompt is designed to guide the robust commercial VLM, GPT-4o, to generate various question-answer pairs related to each map sample. The prompt is designed as: ‘You are an expert in cartography, and you are being tasked to ask questions about the given map. Please generate a set of questions and corresponding answers based on the geographical objects and spatial relationships shown on the map. Ensure the questions are varied and cover different aspects of the map. Do not make up or fabricate any details’.

Subsequently, to ensure the quality of the question-answer data, the initially generated data are manually reviewed and optimized by two experts following similar criteria in collecting MapEval-DB. Ultimately, a high-quality visual question-answering benchmark is collected, which contains 70 base maps and 500 question-answer pairs (400 for urban areas and 100 for rural areas) highly relevant to corresponding base maps. For a comprehensive evaluation of the model’s ability in different visual question-answering tasks on maps, these question-answer pairs are classified into four types:

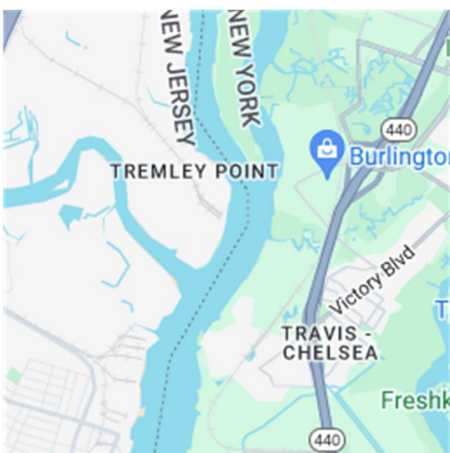
- Geographic object identification: This category focuses on the identification of specific geographic object presented on the map. For example, questions like ‘What is the name of the beach shown in the map?’ aim to identify specific landmarks or geographical entities (Figure 10(a)).
- Spatial relationship recognition: Questions in this category explore the spatial relationships between geographic entities, such as the relative position or distance between them. An example question is ‘Which direction is New York located relative to New Jersey on the map?’ (Figure 10(b)).
- Map symbol identification: This category involves recognizing and interpreting symbols used on the map. For instance, questions like ‘What type of establishment is “Rumba Cubana”?’ assess the model’s proficiency in decoding symbolic representations (Figure 10(c)).
- Content existence verification: This category verifies the presence or absence of certain objects or relationships on the map. An example is ‘Is there a bridge shown in the map?’ (Figure 10(d)).

The distribution of question-answer pairs across different task categories is shown in Table 1. This benchmark can be utilized to comprehensively assess the



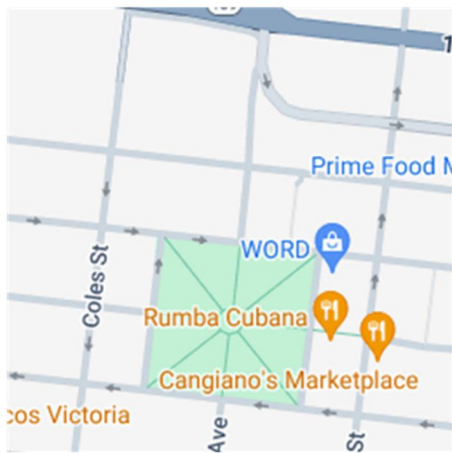
Question: What is the name of the beach shown in the map?
Answer: Orchard Beach.

(a) Geographic object identification



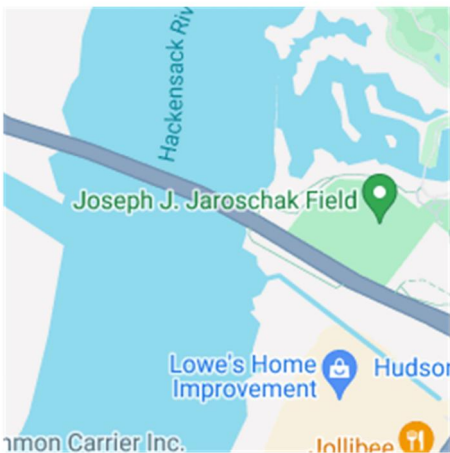
Question: Which direction is New York located relative to New Jersey on the map?
Answer: East

(b) Spatial relationship recognition



Question: What type of establishment is "Rumba Cubana"?"
Answer: Restaurant

(c) Map symbol identification



Question: Is there a bridge shown in the map?
Answer: Yes.

(d) Content existence verification

Figure 10. Illustration of question-answer pair samples in MapEval-QA. All question-answer pairs are classified into four categories: (a) geographic object identification, (b) spatial relationship recognition, (c) map symbol identification, and (d) content existence verification.

Table 1. Quantity distribution of question-answer pairs across different task categories.

Task type	Urban areas	Rural areas
Geographic object identification	160	31
Spatial relationship recognition	145	6
Map symbol identification	73	32
Content existence verification	22	31
Total	400	100

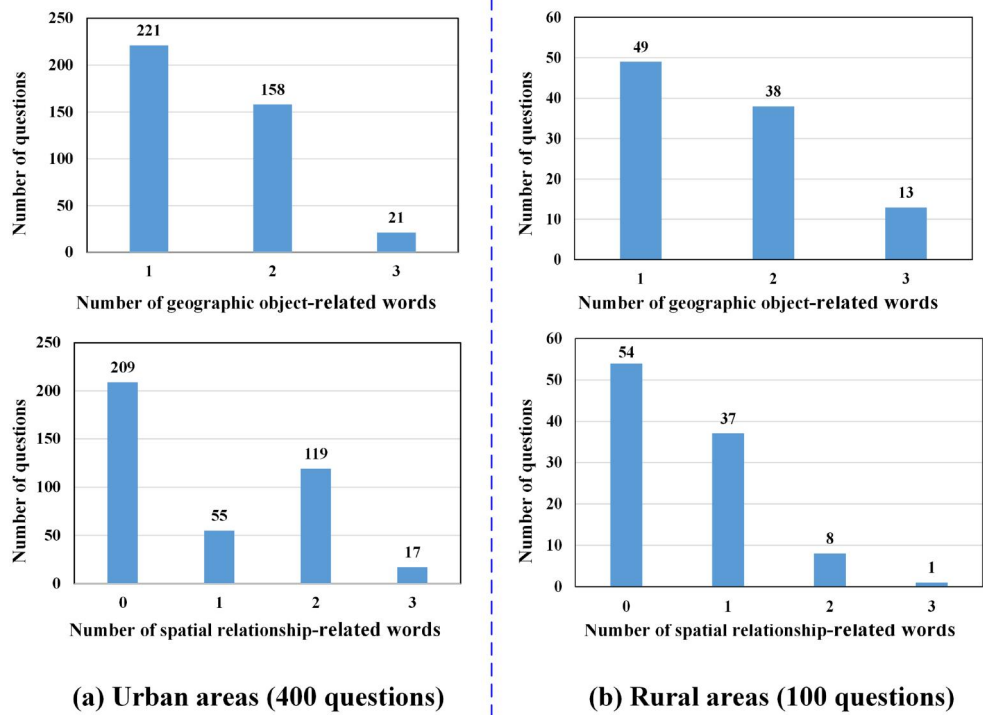


Figure 11. The quantity distributions of geographic object-related words and spatial relationship-related words contained in question-answer pairs for (a) urban areas and (b) rural areas of MapEval-QA.

model’s performance in understanding map information and verify its ability to answer map-related questions. Relevant statistics of MapEval-QA are presented in Figures 11 and 12.

4. Training

With the training data collected, this section elaborates on the training process of MapReader. The framework is presented in Figure 13. Generally, training a multimodal model from scratch requires significant computational resources and a large amount of training data. To address this challenge, the fine-tuning strategy is commonly used, which involves pretraining a foundation model on data from various domains and then further fine-tuning it on domain-specific data for specialized tasks. This strategy

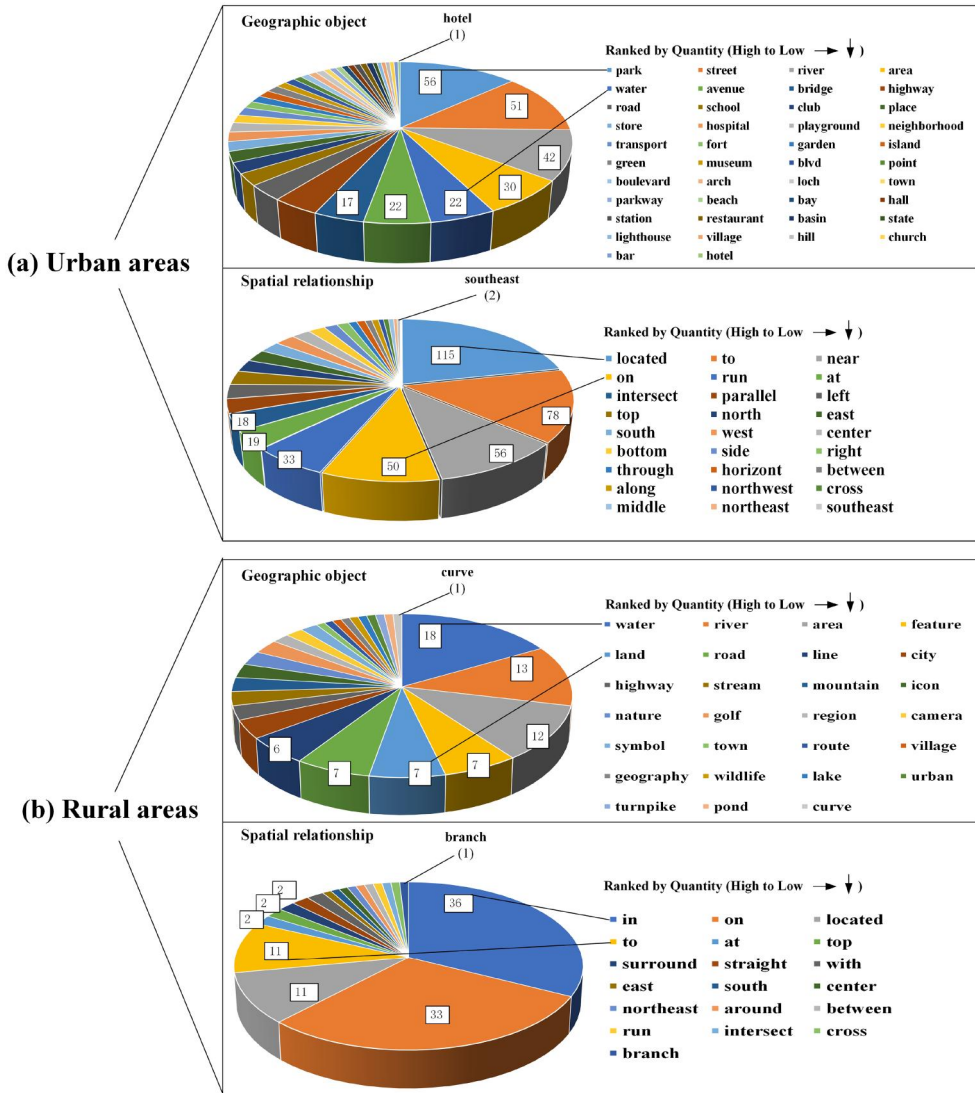


Figure 12. Frequency distribution of geographic object-related and spatial relationship-related words in question-answer pairs for (a) urban areas and (b) rural areas of MapEval-QA.

significantly reduces the computational resource requirements for training a professional model while allowing the fine-tuned model to benefit from the pretraining stage. However, for large models, such as LLMs (e.g. LLaMA-2-7B (Touvron *et al.* 2023) with 7 billion parameters) and VLMs (e.g. CogVLM (Wang *et al.* 2023) with 17 billion parameters), fine-tuning all parameters for specific tasks is still computationally intensive. Therefore, Parameter-Efficient Fine-Tuning (PEFT) is necessary. In this paper, MapReader is trained on top of the general-domain VLM, CogVLM, using a PEFT strategy. Specifically, Section 4.1 introduces the PEFT strategy, and Section 4.2 elaborates on how this strategy is applied to train MapReader.

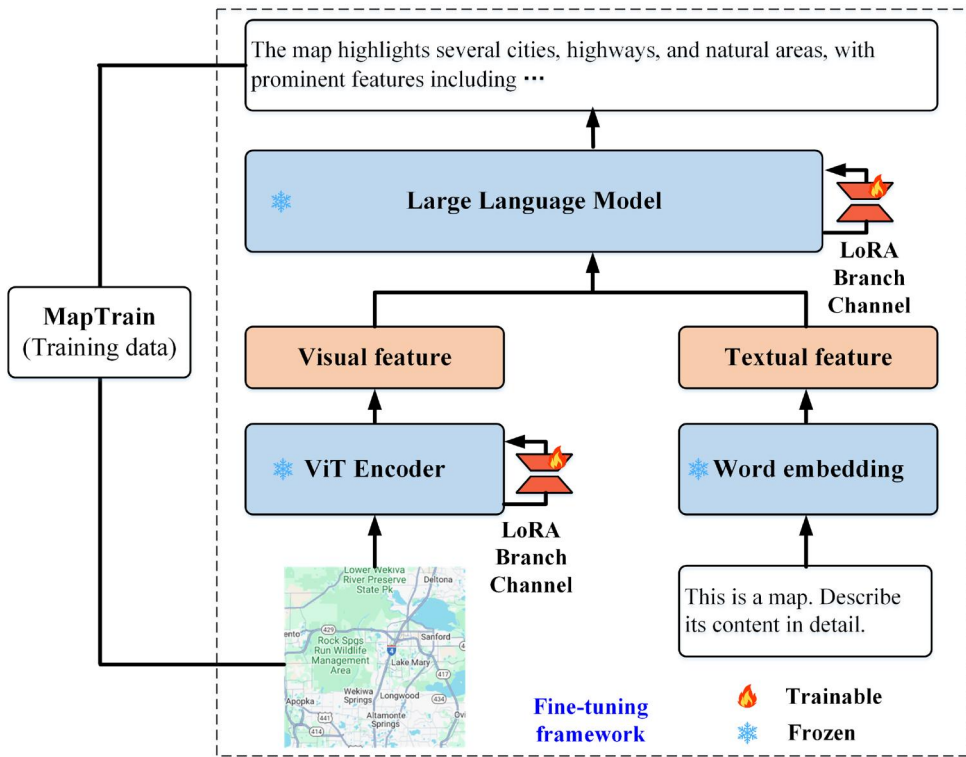


Figure 13. The overall architecture of MapReader.

4.1. Parameter-efficient fine-tuning

Since the emergence of large-scale VLMs, the high cost of computing resources has made fine-tuning all their parameters less feasible. Therefore, there is a need for PEFT methods for fine-tuning large-scale VLMs. These methods aim to reduce computational and storage costs by optimizing a small portion of the model's parameters while achieving performance comparable to fine-tuning all the parameters. Many PEFT methods have been proposed, and each of them is designed for a specific target (Houlsby *et al.* 2019, Pfeiffer *et al.* 2020, Hu *et al.* 2021, Karimi Mahabadi *et al.* 2021). Among these methods, LoRA (Hu *et al.* 2021) is one of the most used PEFT methods and has been validated in fine-tuning both LLMs and VLMs for professional domains. Therefore, in this paper, LoRA is also used to fine-tune MapReader on the top of CogVLM.

LoRA adopts a strategy of freezing parameters of the foundation model and designing an additional trainable branch channel to specifically fit downstream tasks. Specifically, assume the parameter weight matrix of the foundation model to be $W \in R^{d \times k}$, where d and k represent the dimensions of the input and output features, respectively. During the fine-tuning process, W will be frozen with no gradient propagation information, which can save computing and storage consumption. Then, to adapt the foundation model for downstream tasks, an additional branch channel is designed, including two weight matrixes $A \in R^{d \times r}$ and $B \in R^{r \times k}$, where rank r is a

hyper-parameter, and $r \ll \min(d, k)$. Therefore, given an input feature $x \in \mathbb{R}^{n \times k}$, the output hidden feature h can be obtained as follows

$$h = xW + xAB \quad (1)$$

In this way, only the parameters of A and B are trained, which can significantly save training resources. Moreover, for different tasks, adding the branch channel (i.e. A and B) to which part (i.e. W) of the whole network structure and the value of rank r can be customized to accommodate various computational resource constraints. The effectiveness of LoRA has been validated in related work, demonstrating that it can achieve model quality comparable to, or even better than, traditional fine-tuning (Zhang *et al.* 2024d). Consequently, we adopted LoRA in our research.

4.2. The training process of MapReader

In this paper, MapReader is trained on the top of CogVLM. CogVLM is selected majorly considering its state-of-the-art performance on multiple cross-modal benchmarks and tasks, such as image describing and visual question answering. Specifically, the CogVLM model consists of two main components: a Vision Transformer (ViT) encoder and a pretrained LLM (Figure 13). The ViT is used to accept and process the image data into visual features. The input text will be first embedded as textual features that can be understood by models. In CogVLM, a visual expert module is designed for visual-textual feature alignment, and readers can refer to (Wang *et al.* 2023) for more details. After the feature alignment process, the concatenated features will be input a LLM for outputting the final answers. In CogVLM, the EVA2-CLIP-E encoder (Sun *et al.* 2023) is used as the ViT, and Vicuna1.5-7B (Chiang *et al.* 2023) is used as the LLM.

Since our focus is on adapting CogVLM for map analysis, LoRA is applied to both the ViT and LLM. The additional branch channels in the ViT are designed to help CogVLM understand map styles, while those in the LLM are intended to learn the professional map description style (Figure 13). Specifically, LoRA is applied to each transformer layer in the ViT encoder, as well as to the Q, K, and V matrices and the multi-head self-attention layers in LLM (the ‘LoRA branch channel’ in Figure 13). For more details on the Q, K, and V matrices, readers can refer to the Transformer (Vaswani 2017). This strategy effectively reduces the number of parameters required for training and lowers the computational cost. The training process is based on the high-quality map-description pair dataset, MapTrain. Finally, the hyperparameter settings for the training process are as follows: global batch size of 8, learning rate of $1e-5$, 1 training epoch, and a maximum target length of 2,048 tokens. The pretraining process was conducted on $4 \times$ A100 GPUs. As for the LoRA settings, we set the LoRA rank r to 8.

5. Evaluation

In this section, the performance of our trained MapReader on the collected benchmark MapEval is presented. Specifically, there are two tasks to evaluate the model, including map content describing (MpaEval-DB) and map visual question-answering (MapEval-QA). Specifically, the task of map content describing is used to test whether a model can describe the core geographic objects as well as their corresponding spatial

relationships in detail (Section 5.1), and the task of map visual question answering is used to evaluate the model's understanding and responses ability to answer users' questions based on specific map content (Section 5.2). In this paper, four robust general-domain open-source VLMs are used as the baselines, including BLIP2 (Li *et al.* 2023), InstructBLIP (Dai *et al.* 2023), MiniGPT4 (Zhu *et al.* 2023), and the original CogVLM. Moreover, though GPT-4o is used for the initial map description generation, the final map descriptions are further improved with expert refinement. Therefore, the performance of GPT-4o is also presented to show how well it can approximate the ground truth. Finally, several cases are presented in Section 5.3 to show how MapReader can be used to assist map analysis.

5.1. Map content describing

5.1.1. Evaluation metrics

In this task, all models are required to describe the content of given maps in detail. Specifically, for each test map sample in MapEval-DB, a text prompt stating 'This is a map. Describe its content in detail'. is provided along with the map. This input is then processed by MapReader and other baseline models, with the output being the corresponding description (see Figure 13). The generated map descriptions are subsequently evaluated based on two types of evaluation metrics.

5.1.1.1. Rule-based metrics. Given the similarity between map content description and image captioning, three common metrics for evaluating image captions are also applied in this paper. Specifically, these metrics include the BiLingual Evaluation Understudy (BLEU) (Papineni *et al.* 2002), the Metric for Evaluation of Translation with Explicit ORdering (METEOR) (Banerjee and Lavie 2005), and the Consensus-based Image Description Evaluation (CIDEr) (Vedantam *et al.* 2015). These metrics measure the similarity between generated map descriptions and ground-truth labels, with higher scores indicating better alignment with human references. While BLEU emphasizes n-gram matching precision, METEOR accounts for linguistic variations, and CIDEr assesses consensus with multiple references using TF-IDF weighting. Since each metric has different characteristics and emphases, all are employed for a comprehensive evaluation. In this paper, BLEU scores are calculated with n-gram precision for $n = 1, 2, 3$, and 4.

5.1.1.2. Expert evaluation. Besides these rule-based metrics, we also use expert evaluation for a comprehensive evaluation. Accurate identification of core geographic objects and their spatial relationships within maps is essential for effective map analysis. Therefore, we count the number of accurately identified geographic objects and spatial relationships in each test map sample. Figure 14 presents an illustrative example to demonstrate this counting process. It should be noted that not all geographic objects and spatial relationships are exhaustively annotated in each map; core content is labeled, but less prominent geographic features or relationships may be omitted. For example, while relationships such as 'A is on the left of B' and 'B is on the left of C' may be recorded, the relationship 'C is on the right of B' might not be explicitly described. Different models may emphasize different aspects, leading to

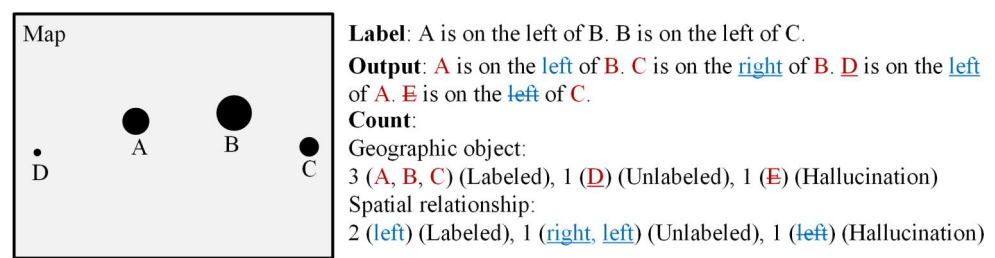


Figure 14. Illustration of the expert evaluation.

Table 2. The evaluation performance of MapReader and baselines on MapEval-DB (rule-based metrics).

	Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	CIDEr
Urban areas	BLIP2	0.00	0.00	0.00	0.00	3.76	0.00
	InstructBLIP	17.91	8.13	2.57	0.82	15.64	1.71
	MiniGPT4	25.16	11.26	4.29	1.56	18.45	2.17
	CogVLM	19.41	10.00	5.29	1.64	17.35	1.62
	GPT-4o	94.57	93.37	92.30	91.52	96.89	747.92
	MapReader	48.89	34.46	25.15	18.90	39.31	74.72
Rural areas	BLIP2	0.01	0.00	0.00	0.00	3.35	0.00
	InstructBLIP	20.86	9.03	1.66	0.44	15.67	4.01
	MiniGPT4	24.38	9.98	2.86	0.52	17.53	5.15
	CogVLM	20.56	9.45	2.87	0.78	16.69	1.10
	GPT-4o	93.26	91.93	90.71	89.79	95.66	756.64
	MapReader	39.33	23.27	13.90	8.66	29.13	35.89

valid descriptions that are absent in the labeled results (e.g. unlabeled content in Figure 14). These cases are counted separately: the numbers of correctly identified geographic objects within and outside the range of labeled results are marked as ‘geographic object’ and ‘geographic object (unlabeled)’, respectively, with the same approach applied to spatial relationships. Additionally, hallucination output in VLMs, which refers to fabricated geographic objects or spatial relationships, is also a significant factor. In this section, hallucinated outputs related to geographic objects and spatial relationships are counted separately as well.

5.1.2. Results

The evaluation results of map descriptions obtained by different models are shown in Tables 2 and 3. First, based on the evaluation performance on rule-based metrics (Table 2), MapReader outperforms all open-source baselines across three evaluation metrics (i.e. BLEU, METEOR, and CIDEr). Furthermore, MapReader achieves consistently better results in both urban and rural areas, demonstrating its effectiveness. GPT-4o performs significantly better than other models on these rule-based metrics. This is because GPT-4o was used to generate initial map descriptions, which were then refined by experts, resulting in outputs that closely match the labels. Since rule-based metrics measure the similarity between model output and ground-truth labels, GPT-4o’s alignment with the labels leads to higher scores. Given that each model emphasizes different aspects and may produce outputs that don’t exactly match the style of the labels, we also include expert evaluation results in Table 3 for a more

Table 3. The evaluation performance of MapReader and baselines on MapEval-DB (expert evaluation).

	Model	Geographic object			Spatial relationship		
		Labeled	Unlabeled	Hallucination	Labeled	Unlabeled	Hallucination
Urban areas	BLIP2	130	2	121	0	0	2
	InstructBLIP	256	5	759	5	19	67
	MiniGPT4	87	8	507	4	13	156
	CogVLM	1,000	76	199	11	28	35
	GPT-4o	1,448	0	84	1,287	0	110
	MapReader	1,188	102	105	344	470	241
Rural areas	BLIP2	30	0	20	0	0	0
	InstructBLIP	43	0	209	6	0	6
	MiniGPT4	27	0	126	5	0	9
	CogVLM	122	5	15	18	16	7
	GPT-4o	161	0	13	147	3	6
	MapReader	130	16	14	41	51	14

Table 4. Quantity statistics of the generated map descriptions by MapReader and baselines.

Model	Total word count	Average word count		Total sentence count	Average sentence count per description
		per description			
BLIP2	1,817	9.1		200	1.0
InstructBLIP	14,726	73.6		719	3.6
MiniGPT4	17,320	86.6		1,005	5.0
CogVLM	13,863	69.3		858	4.3
GPT-4o	24,871	124.4		1,260	6.3
MapReader	24,107	120.5		1,179	5.9

comprehensive comparison. The expert evaluation results indicate that MapReader outperforms all open-source baselines and approaches the performance of GPT-4 in accurately identifying geographic objects and their spatial relationships, further validating the effectiveness of our training data and strategy.

However, the only index where MapReader underperforms compared to other models is in the hallucination of spatial relationships. This issue may arise from CogVLM’s original training on natural images, where spatial relationships differ significantly from those in maps. Addressing this issue may require incorporating more specialized training data and developing relevant alignment strategies, such as reinforcement learning. Nevertheless, MapReader demonstrates superior performance in most cases of map analysis compared to the baselines, confirming the effectiveness of our framework.

Moreover, the word count of map descriptions generated by MapReader and other baseline models is also analyzed. Corresponding word counts of map descriptions obtained by different models are shown in Table 4. The results indicate MapReader generates longer descriptions than the open-source baselines, similar to GPT-4, suggesting that our training data contributes to the increased description length.

5.2. Map visual question answering

5.2.1. Evaluation metrics

Since map users may also focus on only parts of specific maps, the visual question-answering ability of MapReader is also tested on the collected map visual question answering benchmark MapEval-QA. Specifically, there is a question and a

Table 5. The accuracy performance of MapReader and baselines on MapEval-QA.

	Model	Geographic object identification (%)	Spatial relationship recognition (%)	Map symbol identification (%)	Content existence verification (%)	Total (%)
Urban areas	BLIP2	28.8	11.0	37.0	13.6	23.0
	InstructBLIP	31.9	21.4	41.1	59.1	31.3
	MiniGPT4	13.1	9.7	30.1	77.3	18.5
	CogVLM	83.1	55.9	72.6	86.4	71.5
	GPT-4o	91.3	84.1	93.2	100.0	89.5
	MapReader	86.9	78.6	91.8	90.9	85.0
Rural areas	BLIP2	38.7	33.3	40.6	32.3	37.0
	InstructBLIP	32.3	33.3	62.5	67.7	53.0
	MiniGPT4	9.7	33.3	50.0	48.4	36.0
	CogVLM	71.0	50.0	75.0	87.1	76.0
	GPT-4o	83.9	83.3	90.6	90.3	88.0
	MapReader	83.9	66.7	78.1	93.5	84.0

corresponding map in each test sample of MapEval-QA. The question is related to map content, and all the models are required to answer the question based on the map. To present the evaluation on a finer scale, all the questions are classified into four types (Table 1).

To effectively evaluate the visual question-answering performance of different models, each model’s output for a given question is compared with the corresponding ground-truth label. Finally, accuracy is calculated as the ratio of correctly answered questions to the total number of questions. This metric provides a measure of each model’s capability in map understanding and visual question-answering.

5.2.2. Results

All the results are listed in Table 5. From these results, MapReader outperforms all open-source baselines and approaches the performance of GPT-4o. These findings validate the effectiveness of our framework. However, MapReader performs worse on spatial relationship recognition-related questions compared to its performance on other types of questions. This pattern is consistent with its performance in the map description task. Therefore, our future work will focus on improving MapReader’s ability to understand spatial relationships.

5.3. Case studies

In this section, two types of cases, including map description and map visual question answering, are presented in Figures 15–17. In the first case shown in Figure 15, MapReader accurately identifies most geographic objects and their corresponding spatial relationships, yielding results similar to the ground truth label. This outcome validates the effectiveness of our framework, demonstrating that MapReader can learn to understand map content with professional training. This conclusion is further supported by comparing our result with that of CogVLM, as MapReader is trained on top of CogVLM. While CogVLM can identify some geographic objects, its descriptions are simpler and include some hallucinated content. Moreover, by comparing the results of MapReader with GPT-4, one of the most robust commercial VLMs, we find that

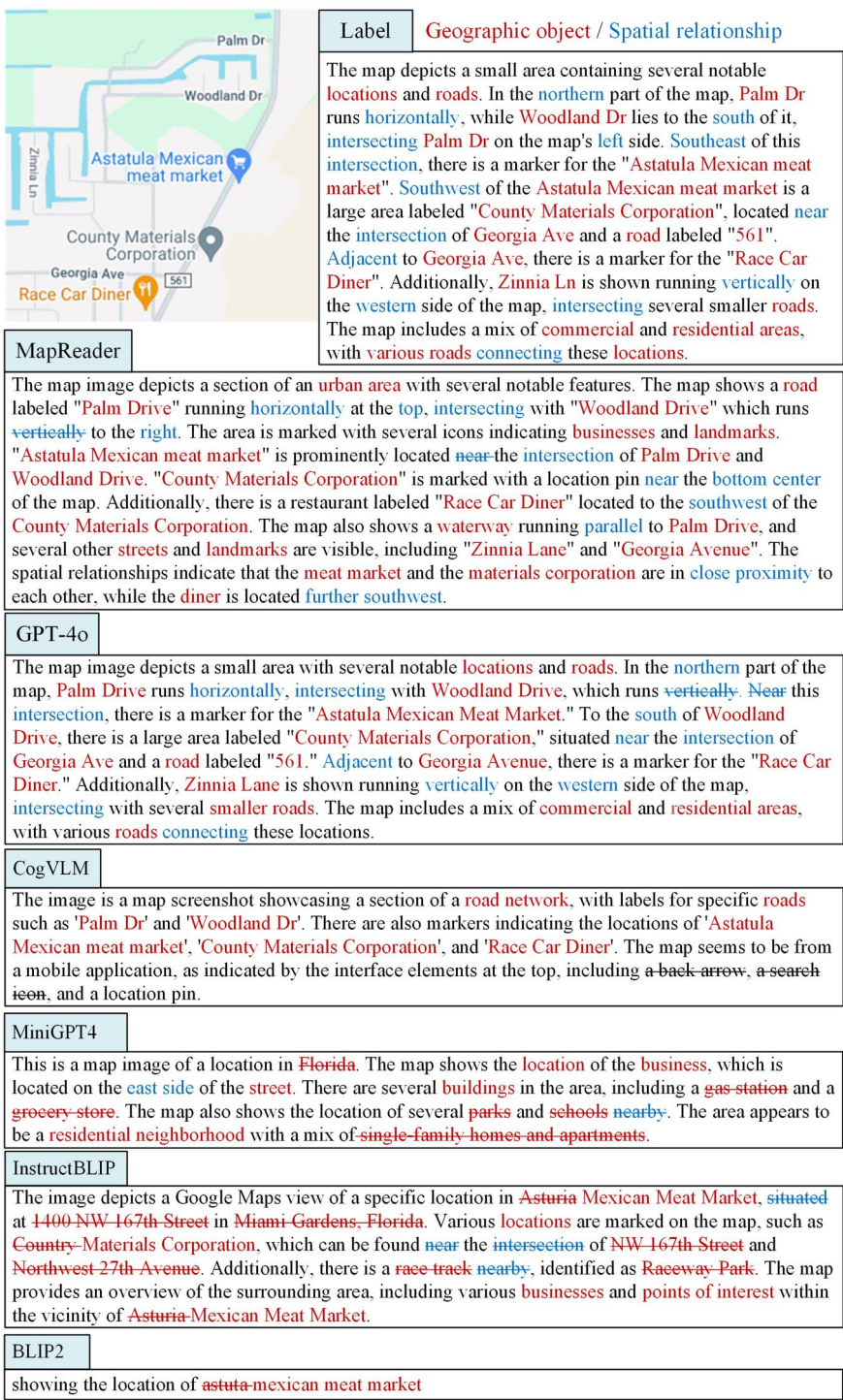


Figure 15. Map description case 1: red text indicates identified geographic objects, blue text indicates identified spatial relationships, and strikethroughs represent misidentified content.

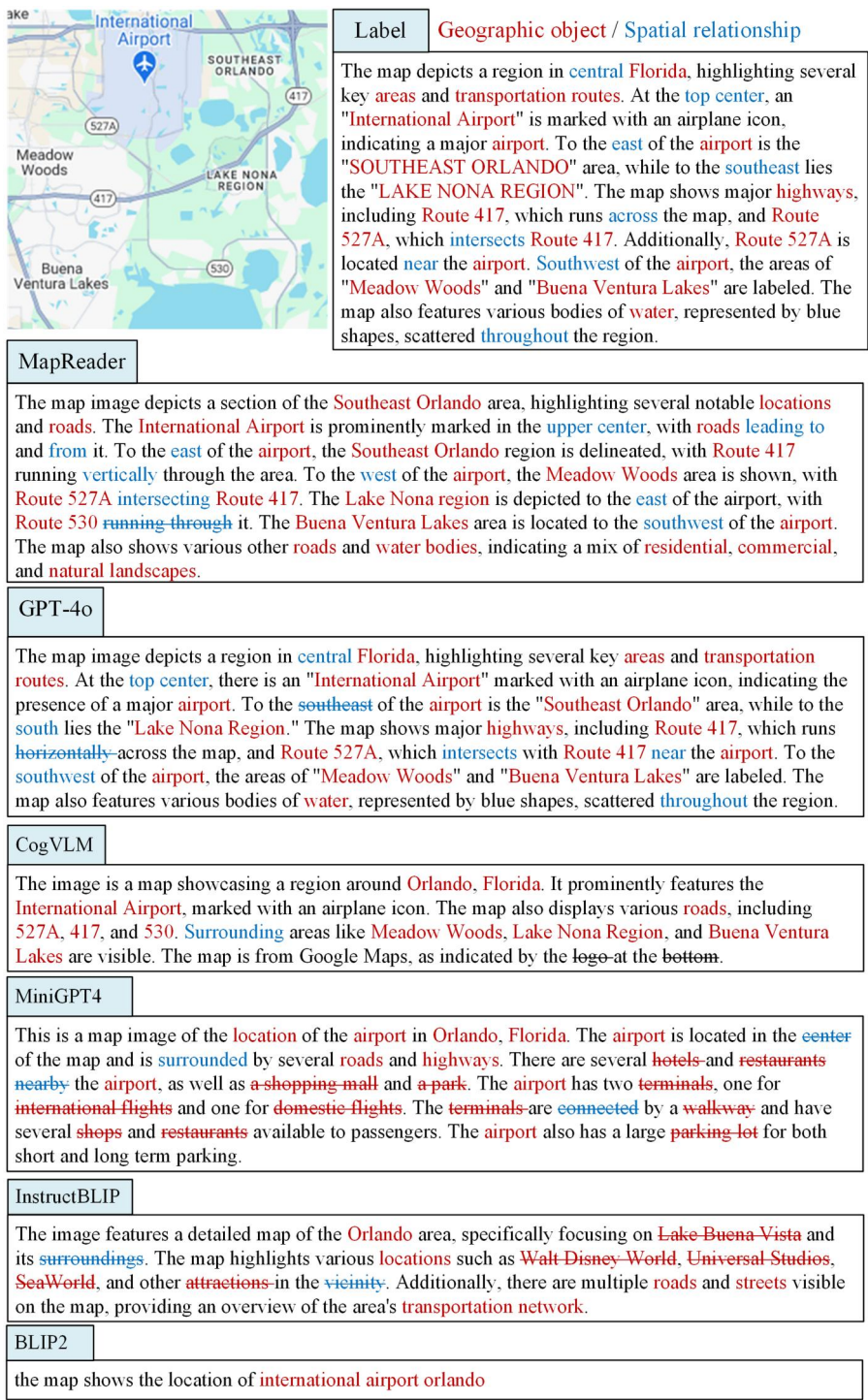
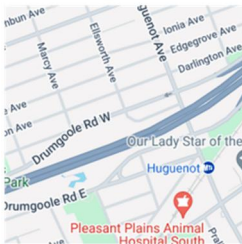


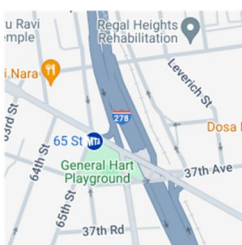
Figure 16. Map description case 2: red text indicates identified geographic objects, blue text indicates identified spatial relationships, and strikethroughs represent misidentified content.

Case 1



Question: Which avenue intersects with Drumgoole Rd W and Drumgoole Rd E?
Answer: Huguenot Ave
MapReader: Huguenot Ave intersects with Drumgoole Rd W and Drumgoole Rd E. ✓
GPT-4o: Huguenot Ave ✓
CogVLM: Ionia Ave intersects with Drumgoole Rd W and Drumgoole Rd E. ✗
MiniGPT4: The intersection of Drumgoole Rd W and Drumgoole Rd E is located in the city of Los Angeles, California. ✗
InstructBLIP: Dundas St W ✗
BLIP2: st john's ave ✗

Case 2



Question: What is the name of the restaurant shown on the map?
Answer: i Nara
MapReader: i Nara ✓
GPT-4o: Dosa Delight ✗
CogVLM: Bhoomi Nara ✗
MiniGPT4: The name of the restaurant shown on the map is not specified. ✗
InstructBLIP: The name of the restaurant shown on the map is General Hart Playground. ✗
BLIP2: san francisco, california - san francisco, california - ✗

Figure 17. Two cases of Visual Question Answering (VQA) on maps.

MapReader can identify a waterway on the map that GPT-4 missed, showcasing MapReader’s professional map understanding ability. Considering other baselines, their performance is still unsatisfactory because they are trained on natural images, which present intrinsic differences compared to maps. This also validates the necessity of training a specialized VLM for map analysis. The second case presented in Figure 16 exhibits a similar phenomenon.

Following this, two cases of visual question answering on maps are also presented in Figure 17. The first question asks for a geographic object with a specific relationship to other map content. In practice, this question is somewhat challenging for human due to the presence of many roads and the complex layout of the map. Moreover, only part of the answer is visible (i.e. the road name ‘___guenot Ave’). However, MapReader is able to take a global view of the map, identify another relevant geographic object (‘Huguenot’), and reason from these clues to produce the correct answer (‘Huguenot Ave’). In the second case, MapReader understands the semantics of ‘restaurant’ and correctly outputs the answer. Both cases validate the effectiveness of MapReader.

6. Discussion

6.1. The impact of different selections for the hyperparameter r in LoRA

In this section, we investigate the impact of various selections for the hyperparameter r in LoRA on the performance of MapReader. Specifically, larger values of r correspond to more fine-tuned parameters. For a comprehensive assessment, we trained MapReader with commonly used values of r , with the results presented in Table 6. The results indicate that MapReader achieves optimal performance with relatively small values of r (e.g. 4 and 8). Moreover, while performance varies with different selections of r , all results exceed those of baselines with similar parameter sizes (Table 2), further validating the robustness of our framework.

Table 6. The evaluation performance of MapReader on MapEval-DB with different selections for the hyperparameter r in LoRA.

r	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	CIDEr
4	48.82	34.08	24.73	18.56	39.52	82.52
8	48.89	34.46	25.15	18.90	39.31	74.72
16	48.68	34.09	24.78	18.55	38.62	71.78
32	48.66	33.98	24.59	18.38	38.52	78.87

Note: Bold values are used to indicate the best accuracy achieved under the current configuration.

Table 7. The evaluation performance of MapReader to process maps with varying scales.

Map scale	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	CIDEr
10	46.66	28.68	19.39	13.22	34.23	45.17
11	41.90	27.03	18.94	13.59	32.24	58.19
12	46.63	32.47	22.73	16.17	36.02	23.68
13	43.08	27.46	18.85	13.58	32.66	6.63
14	50.63	34.38	24.62	18.51	38.67	101.53
15	49.24	34.98	25.50	19.20	39.51	76.26
16	49.74	35.77	26.49	20.11	40.87	78.97

Note: Bold values are used to indicate the best accuracy achieved under the current configuration.

6.2. The ability of MapReader to process maps with varying scales

Since map scale is an important map element, this section investigates MapReader's ability to process maps at varying scales. The experimental results are listed in Table 7. From these results, MapReader achieves the best performance on rule-based metrics with maps at scales of 14 and 16. For BLEU and METEOR metrics, MapReader demonstrates relatively stable performance across maps of different scales. A possible reason is that maps of different scales are often generalized based on expert knowledge (i.e. map generalization (Yu *et al.* 2020)), ensuring an appropriate number of geographic objects to avoid overlap or clutter. Moreover, the results for the CIDEr metric indicate that MapReader performs worse on maps at scale 13, possibly due to an insufficient number of maps at this scale in the training data (Figure 4). Collecting more maps at this scale is a priority for future work.

6.3. The ability of MapReader to process maps across different regions

Since our training data were collected from the USA, this section examines MapReader's ability to process maps from different regions. Specifically, we collected additional evaluation maps from three other countries: China (10 maps), North Korea (10 maps), and the United Kingdom (10 maps). All the map samples are available at this link: <https://github.com/AGI-GIS/MapReader>. The results are presented in Table 8. From the results, MapReader consistently outperforms the open-source models across all these regions, which supports the effectiveness of our method.

6.4. Limitations

In this section, the limitations of this study are discussed.

Table 8. The evaluation performance of MapReader across different regions.

	Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	CIDEr
China	BLIP2	0.01	0.00	0.00	0.00	3.57	0.00
	InstructBLIP	22.75	9.40	3.53	0.00	17.62	2.04
	MiniGPT4	30.79	13.61	5.91	2.91	21.89	1.12
	CogVLM	20.78	10.38	4.17	0.88	17.50	1.18
	GPT-4o	95.71	94.58	93.56	92.81	98.15	809.31
	MapReader	43.86	27.88	18.73	13.26	34.76	25.65
North Korea	BLIP2	0.02	0.00	0.00	0.00	3.08	0.00
	InstructBLIP	19.80	7.88	2.01	0.00	15.83	7.16
	MiniGPT4	27.79	12.99	4.21	1.54	19.71	6.01
	CogVLM	20.57	10.49	5.02	2.07	17.44	0.00
	GPT-4o	96.72	96.38	96.03	95.68	98.86	766.54
	MapReader	42.71	26.44	16.30	10.06	30.87	65.32
United Kingdom	BLIP2	0.00	0.00	0.00	0.00	3.02	0.00
	InstructBLIP	14.46	6.62	2.71	1.28	15.92	6.71
	MiniGPT4	25.95	12.70	3.25	1.76	19.72	2.02
	CogVLM	20.28	11.15	5.64	0.47	17.54	0.61
	GPT-4o	98.21	97.59	96.96	96.37	98.24	947.13
	MapReader	46.79	30.42	20.29	14.09	38.42	52.74

6.4.1. Fine-grained geographic object descriptions

In this paper, we proposed a strategy that combines self-instruction with expert refinement for map description label generation. The experimental results validate the effectiveness of this strategy, which can greatly reduce manual labeling effort. However, although MapReader has robust performance in accurately identifying geographic objects and the corresponding spatial relationships between them, its ability to further identify finer-scale information about geographic objects still needs improvement. For example, in [Figure 15](#), there is a road labeled '561' in this map. Though MapReader accurately identifies this road, it does not recognize it as a county road. Therefore, we believe that further improving MapReader's ability to provide finer-scale information about geographic objects is important. One possible solution to this problem is collecting fine-grained geographic object descriptions, which is a focus of our future work.

6.4.2. Expanding support for multilingual and diverse map styles

Although the experimental results show that MapReader can still outperform open-source models across different regions, it should still be noted that MapReader cannot support any languages besides English. An illustrative example is presented in [Figure 18](#). From the map, we observe that there are also Chinese labels for geographic objects. However, MapReader cannot output the corresponding information since the language model in MapReader only supports English. Therefore, expanding multilingual support is also a focus of our future work. Moreover, our study exclusively uses maps from Google Maps for training. However, the map data from a single service may be incomplete in certain areas. To address this limitation, we plan to include maps with diverse styles in the training data. This enhancement will enable our model to perform effectively in regions where Google Maps is unavailable, leveraging data from alternative map services.

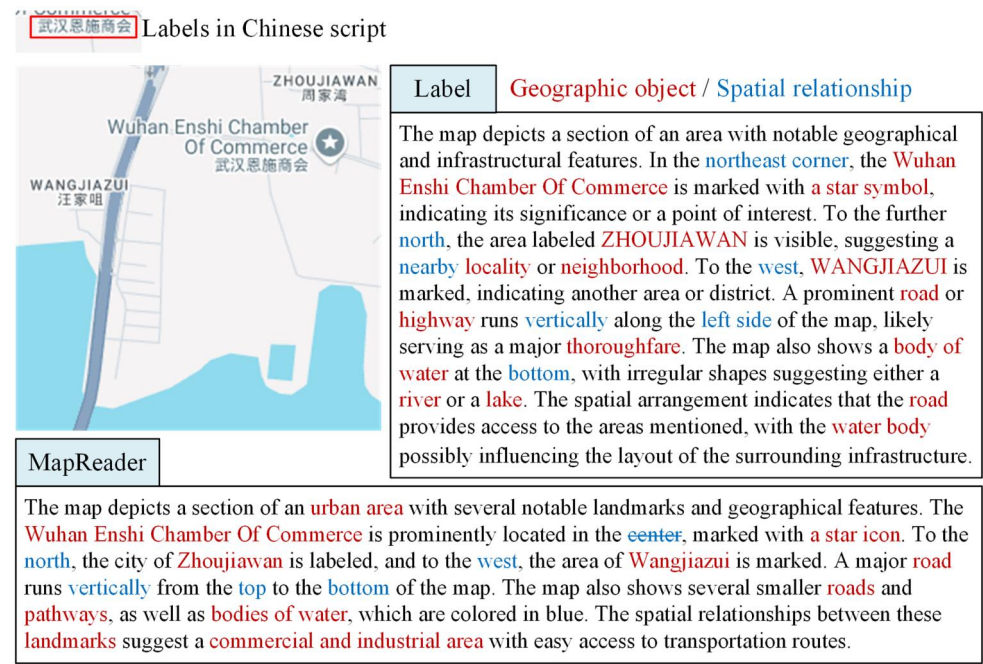


Figure 18. Presentation of a map from China. This map contains labels in Chinese, but MapReader currently supports only English.

7. Conclusion

In this paper, we propose a framework for collecting a map-description pair dataset, called MapTrain, designed to fine-tune a professional VLM for map analysis. The dataset is collected using a strategy that combines self-instruct with expert refinement. Specifically, initial map descriptions are generated by a robust commercial VLM, GPT-4o, and are subsequently reviewed and optimized by cartographic experts. This approach allows the creation of MapTrain with minimal human labor. We then fine-tune a professional VLM, MapReader, using MapTrain with a lightweight Low-Rank Adaptation (LoRA) fine-tuning method. Moreover, to evaluate the performance of VLMs in map analysis, we also develop a benchmark, MapEval, from two perspectives: testing map description capabilities (MapEval-DB) and map visual question answering (MapEval-QA). Experimental results show that MapReader significantly outperforms open-source baselines on MapEval in most cases and achieves performance closely matching that of GPT-4o, which is one of the most robust commercial VLMs. These results validate the effectiveness of our framework, and our collected datasets can also serve as a foundational resource for advancing professional VLMs in the cartography community.

However, although MapReader outperforms the baselines, there are still areas for improvement. First, the experimental results show that MapReader generates more hallucinations related to spatial relationships than the baselines, despite achieving better performance in correctly identifying spatial relationships and producing longer, more detailed map descriptions. We believe this issue arises because MapReader is built on a general-domain VLM that was trained on a large number of natural images, where spatial

relationships differ significantly from those in maps. Therefore, collecting more specialized map data to train a foundational cartographic VLM may help address this problem. Additionally, while MapReader has a smaller parameter size compared to commercial and some open-source VLMs, it is also important to develop professional VLMs with even fewer parameters, enabling deployment on personal devices like mobile phones. To achieve this, we plan to explore strategies such as quantization and distillation to reduce MapReader's parameter size while maintaining its performance. Finally, while the human evaluation strategy (Figure 14) we employed supplements the evaluation of rule-based metrics, the process still requires expert knowledge, which may restrict the framework's applicability to other proposals. Therefore, further development of an automatic evaluation framework would greatly enhance the reproducibility and comparability of results across different proposals in this research area.

Acknowledgements

The authors extend their sincere gratitude to the anonymous reviewers and the editor for their time, expertise, and constructive comments. Their insights and feedback have significantly enhanced the content of this article.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The project was supported by Natural Science Foundation of Hubei Province, China [2024AFD412], National Natural Science Foundation of China [42371446 and 42071442], and by the Fundamental Research Funds for National Universities, China University of Geosciences [No. 2024XLA17]. This research was also supported by Meituan. Yifan Zhang acknowledges the support of the China Scholarship Council program (Project ID: 202306410085).

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Data and codes availability statement

The data and code supporting the findings of this study are available at this link <https://github.com/AGI-GIS/MapReader>. Additionally, in this link, we have uploaded a visualization video to demonstrate how MapReader works, providing readers with a deeper understanding of MapReader.

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Appendix A. The presentation of maps with different scales

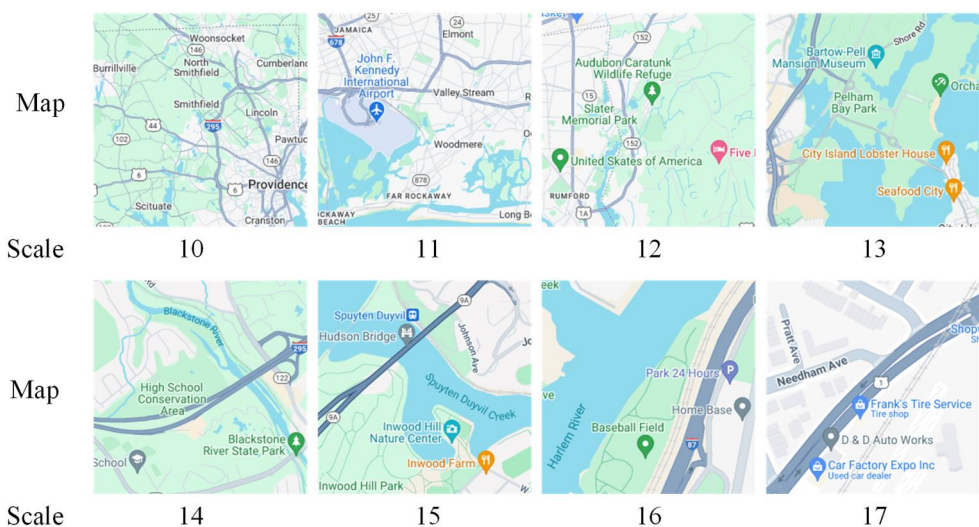


Figure A1. The presentation of maps with different scales.